Online Appendix (E-Companion) to "Who Pays for the Next Attempt? Contract Design and Escalation Rules for AI-Assisted Software Maintenance"

This online appendix contains the survival function and estimation details (Appendices A and B), reproducibility notes (Appendix C), the proofs (Appendix D), and additional numerical and empirical results (Appendices E to H). Section numbers refer to the manuscript.

Appendix A. Survival Function and Posterior Predictive (Beta Family)

Given p, $P(T > j \mid p) = (1-p)^j$. Integrating over $p \sim \text{Beta}(a, b)$ gives $S(j) = B(a, b+j)/B(a, b)$. After n failures the posterior is Beta(a, b+n), whose mean is $a/(a+b+n) = q_n$.

Appendix B. Estimation

With counts (n samples, k successes) per defect, the Beta-binomial likelihood is $C(n,k)\cdot B(k+a, n-k+b)/B(a,b)$. First-success data censored at K attempts contribute $B(a+1, b+t-1)/B(a,b)$ for a defect solved at attempt t and $B(a, b+K)/B(a,b)$ for an unsolved one. Both likelihoods are maximized over (log a, log b) with the Nelder–Mead algorithm.

Appendix C. Reproducibility

The replication code file accompanying the submission contains the code for the contract equilibria (`contracts.py`, `robustness.py`), private quality (`asymmetric.py`, `asym_sweep.py`), competition (`diff_bertrand.py`, `diff_sweep.py`), proof checks (`verify_proofs.py`) and distribution-free checks of Propositions 1, 5, and 6 (`general_theory.py`); the empirical analyses of RepairBench (`load_repairbench.py`, `analyze.py`), the SWE-bench logs (`swebench_repo_level.py`) and the HumanEvalFix sampling and analysis (`sample_humanevalfix.py`, `analyze_hef.py`); and the cost calibration and sensitivity analysis (`calibration.py`, `sensitivity.py`, `fig_sensitivity.py`).

Appendix D. Proofs

Lemma A (steps). Under contract B, $N_B(f) = n \geq 1$ if and only if $q_n(H-f) \leq v < q_{n-1}(H-f)$, that is, $f \in [\varphi_n, \varphi_{n-1})$ with $\varphi_n = H - v/q_n$. For the Beta family $\varphi_n = H - v(n+a+b)/a$, so the steps have equal width v/a. Also $N_B(f) = 0$ if and only if $f \geq \varphi_0$. At $f = \varphi_n$ the firm is indifferent about attempt n+1 and stops.

Lemma B (comparison across a step boundary). Let $L(n) = \varphi_n P(n) - kE(n)$ and $R(n) = \varphi_{n-1}P(n) - kE(n)$. Then $R(n) - L(n-1) = \varphi_{n-1}[P(n) - P(n-1)] - k[E(n) - E(n-1)] = S(n-1)[\varphi_{n-1}q_{n-1} - k] = S(n-1)[q_{n-1}H - (k+v)]$, using $P(n) - P(n-1) = q_{n-1}S(n-1)$, $E(n) - E(n-1) = S(n-1)$, and $q_{n-1}(H - \varphi_{n-1}) = v$. The sign of this expression is the sign of the efficiency of attempt n, which is equivalent to the sign of $\varphi_{n-1} - f^e$.

Proof of Proposition 4. Profit within step n is $fP(n) - kE(n)$, which has slope $P(n) > 0$. Hence $L(n) < \Pi < R(n)$ in the interior of the step, and $R(n) - L(n) = (\varphi_{n-1} - \varphi_n)P(n) > 0$. Let $n_0 = N^{FB} \geq 1$, so that $f^e \in [\varphi_{n_0}, \varphi_{n_0-1})$. For (b), if $n > n_0$ then $\varphi_{n-1} \leq \varphi_{n_0} \leq f^e$, so Lemma B gives $R(n) \leq L(n-1) < R(n-1)$. Thus R is strictly decreasing from n_0 onward and $\hat{n} \leq n_0$. For (a), take $f' \leq f^e$ with $n = N_B(f') \geq n_0$. If $n = n_0$, any $f \in (f^e, \varphi_{n_0-1})$ has $\Pi(f) > \Pi(f')$ by monotonicity within the step. If $n > n_0$, then $\Pi(f') < R(n) < R(n_0)$, and the supremum of Π on (f^e, φ_{n_0-1}) equals $R(n_0)$. ∎

Proof of Proposition 5. (a) Each term of G_{min} is nonnegative because $q_j \geq q_{N-1}$ for $j < N$. For $N = 1$, $P(1)/q_0 - E(1) = 1 - 1 = 0$. When N increases to N+1, the existing terms weakly rise because $q_N \leq q_{N-1}$, and the new term is zero. (b) The firm chooses depth N only if $q_{N-1}(H-f) > w+v$, so $f < \varphi_{N-1} - w/q_{N-1}$. Then $\Pi = (w-k)E + fP < (w-k)E + [\varphi_{N-1} - w/q_{N-1}]P = R(N) - w[P/q_{N-1} - E]$. The identity $R(N) = W(N) -$

$G_{min}(N)$ follows from $\varphi_{N-1}P - kE = HP - (k+v)E - v[P/q_{N-1} - E]$. (c) At $(0, \varphi_{N-1})$ participation holds because $vE + \varphi_{N-1}P \le HP$ is equivalent to $q_{N-1}E \le P$, which is true because $P = \Sigma_{j<N} S(j)q_j$. (d) $G_{min}(N+1) - G_{min}(N) = vP(N)(1/q_N - 1/q_{N-1})$ follows from $P(N+1) = P(N) + q_N S(N)$ and $E(N+1) = E(N) + S(N)$, while $W(N+1) - W(N) = S(N)[q_N H - (k+v)]$. ∎

Appendix E. Detailed Sensitivity Results

The table reports, for each regime and heterogeneity profile in Section 5.4, the share of grid points at which AI repair is worth using and the comparison of the two contract forms. Monopoly equilibria are computed exactly from the budget steps.

Regime	Heterogeneity profile	Grid points	AI repair worth using	Welfare: success fee > per-attempt	Vendor profit: success fee ≥ per-attempt	Firm saving: success fee > per-attempt	Median welfare / first best (per-attempt, success fee)	Median N^{FB}, vendor's depth	Vendor's depth below N^{FB}	Rent share of $W(N^{FB})$: median [min–max]
Function-level, automated verification	RepairBench-like (0.12,0.39)	180	100%	97%	100%	97%	0.48, 0.87	80, 13	100%	90% [9-100%]
Function-level, automated verification	SWE-Verified-like (0.25,0.34)	180	100%	100%	100%	100%	0.54, 0.90	166, 18	100%	91% [9-100%]
Function-level, automated verification	HumanEvalFix-Llama-like (0.50,0.58)	180	100%	100%	100%	100%	0.50, 0.93	330, 26	100%	91% [9-100%]
Function-level, manual triage per attempt	RepairBench-like (0.12,0.39)	144	75%	22%	100%	22%	0.91, 0.91	2, 1	61%	67% [0-99%]
Function-level, manual triage per attempt	SWE-Verified-like (0.25,0.34)	144	88%	33%	100%	33%	0.86, 0.87	3, 1	71%	82% [0-100%]
Function-level, manual triage per attempt	HumanEvalFix-Llama-like (0.50,0.58)	144	88%	57%	100%	57%	0.72, 0.88	6, 2	86%	95% [0-99%]
Repository-level agentic	RepairBench-like (0.12,0.	100	100%	76%	100%	76%	0.67, 0.92	9, 4	92%	37% [0-89%]

Regime	Heterogeneity profile	Grid points	AI repair worth using	Welfare: success fee > per-attempt	Vendor profit: success fee ≥ per-attempt	Firm saving: success fee > per-attempt	Median welfare / first best (per-attempt, success fee)	Median N^{FB}, vendor's depth	Vendor's depth below N^{FB}	Rent share of $W(N^{FB})$: median [min–max]
	39)									
Repository-level agentic	SWE-Verified-like (0.25,0.34)	100	100%	85%	100%	85%	0.66, 0.92	19, 6	99%	43% [0-89%]
Repository-level agentic	Human EvalFix-Llama-like (0.50,0.58)	100	100%	95%	100%	95%	0.56, 0.93	37, 9	100%	44% [3-89%]

Appendix F. Calibration of the Cost Parameters

F.1 Calibration script output

Output of `results_calibration.txt`.

```
Loaded hourly cost of a software developer: $64.44/h x 1.418 = $91/h
SWE-bench Verified expected hands-on time (annotated, familiar
engineer): 0.52 h => H = $48; unfamiliar-engineer multiplier 2x-4x =>
H = $96-$191
Bucket values: <15min => $15; 15-60min => $46; 1-4h => $183-$366

Per-attempt vendor cost k for function-level repair (price list x
observed tokens):
  function-level, non-reasoning (RepairBench: ~1,700 in / ~800 out)
     Gemma-3-4B (small)       k = $0.00011
     Qwen3-Coder-30B-A3B      k = $0.00034
     Qwen3-Coder              k = $0.00131
     Claude Sonnet 5.5        k = $0.01140
     o4-mini (reasoning)      k = $0.00539
  function-level, reasoning (RepairBench o4-mini: ~1,400 in / ~10,400
out)
     Gemma-3-4B (small)       k = $0.00058
     Qwen3-Coder-30B-A3B      k = $0.00301
     Qwen3-Coder              k = $0.01083
     Claude Sonnet 5.5        k = $0.10692
     o4-mini (reasoning)      k = $0.04736
Repository-level agentic repair: published cost per task $0.60-$0.84
(one trajectory or one multi-sample pipeline)
Verification cost v per attempt: automated tests only (CI run) ~
$0.01-$0.50; with 2-10 min of manual triage ~ $3-$15 (at $91/h)
```

F.2 Running example (Table 1 and Figure 1)

Output of `results_running_example.txt`.

```
Beta(0.12,0.39); H=100.0, v=1.0, k=0.2; prior mean success 0.235
q_n (success prob of attempt n+1 after n failures): n=0: 0.2353, n=1:
0.0795, n=2: 0.0478, n=3: 0.0342, n=4: 0.0266, n=5: 0.0218, n=9:
0.0126, n=10: 0.0114, n=19: 0.0062, n=29: 0.0041
posterior mean after 10 failures 0.0114; after 30 failures 0.0039
fraction of defects with p<0.01: 0.47 ; p<0.05: 0.57 ; p>0.5: 0.22 ;
p>0.9: 0.10
marginal value q_n*H vs efficient cost k+v = 1.2: attempt 10 (n=9):
1.26; attempt 11 (n=10): 1.14
N | P(N) | E(N) | W(N) | G_min(N) | R(N) | phi_(N-1)
 1 | 0.235 | 1.00 | 22.33 | 0.00 | 22.33 | 95.8
 2 | 0.296 | 1.76 | 27.49 | 1.96 | 25.53 | 87.4
 3 | 0.330 | 2.47 | 30.01 | 4.43 | 25.58 | 79.1
 5 | 0.370 | 3.79 | 32.44 | 10.11 | 22.33 | 62.4
10 | 0.420 | 6.82 | 33.86 | 26.50 | 7.36 | 20.8
20 | 0.467 | 12.38 | 31.83 | 63.52 | -31.69 | -62.6
per-attempt monopoly price q_0 H - v = 22.53; success fee at step 3:
phi_2 = 79.1
human-only per 1,000 defects: $100,000; first best N=10: W=$33,857
```

F.3 Incentive rent (Table 2)

Output of `results_incentive_rent.txt`.

```
Heterogeneity (a,b) | N^FB^ | W(N^FB^) | G_min(N^FB^) | R(N^FB^) |
Rent/W(N^FB^) | vendor-optimal depth n_hat, R(n_hat)
RepairBench-like (0.12,0.39) | 10 | 33.9 | 26.5 | 7.4 | 78% | 3, 25.6
SWE-Verified-like (0.25,0.34) | 21 | 63.5 | 52.2 | 11.3 | 82% | 4,
50.2
Synthetic (0.4,3.0) | 30 | 43.9 | 35.1 | 8.8 | 80% | 10, 30.4
```

Appendix G. Additional Numerical Results

G.1 Monopoly equilibria and random sweep (Table 3 and Section 5.1)

Output of `results_contracts.txt`.

```
RepairBench-like (a=.12,b=.39) | H=100 v=1.0 k=0.2
   first-best: N=  10 surplus=  33.86
   per-attempt (monopoly w*=22.52): N=   1 vendor= 22.32 firm_saving=
0.01 welfare=  22.33 (66% of FB)
   success-fee (monopoly f*=79.04): N=   3 vendor= 25.57 firm_saving=
4.44 welfare=  30.01 (89% of FB)
   success-fee at efficient f=Hk/(k+v)=16.67: N=  10 vendor=  5.64
firm_saving= 28.21 welfare=  33.86

RepairBench-like (a=.12,b=.39) | H=100 v=1.0 k=1.0
   first-best: N=   6 surplus=  29.53
   per-attempt (monopoly w*=22.49): N=   1 vendor= 21.49 firm_saving=
```

```
0.04 welfare=  21.53 (73% of FB)
   success-fee (monopoly f*=87.39): N=   2 vendor= 24.11 firm_saving=
1.97 welfare=  26.08 (88% of FB)
   success-fee at efficient f=Hk/(k+v)=50.00: N=   6 vendor= 14.76
firm_saving= 14.76 welfare=  29.53

RepairBench-like (a=.12,b=.39) | H=300 v=2.0 k=0.5
   first-best: N=  14 surplus= 110.28
   per-attempt (monopoly w*=68.52): N=   1 vendor= 68.02 firm_saving=
0.07 welfare=  68.09 (62% of FB)
   success-fee (monopoly f*=258.13): N=   3 vendor= 83.88 firm_saving=
8.87 welfare=  92.74 (84% of FB)
   success-fee at efficient f=Hk/(k+v)=60.00: N=  14 vendor= 22.06
firm_saving= 88.22 welfare= 110.28

RepairBench-like (a=.12,b=.39) | H=100 v=0.2 k=0.2
   first-best: N=  30 surplus=  42.18
   per-attempt (monopoly w*=23.32): N=   1 vendor= 23.12 firm_saving=
0.01 welfare=  23.13 (55% of FB)
   success-fee (monopoly f*=87.44): N=   8 vendor= 34.25 firm_saving=
3.95 welfare=  38.21 (91% of FB)
   success-fee at efficient f=Hk/(k+v)=50.00: N=  30 vendor= 21.09
firm_saving= 21.09 welfare=  42.18

SWE-Verified-like (a=.25,b=.34) | H=100 v=1.0 k=0.2
   first-best: N=  21 surplus=  63.53
   per-attempt (monopoly w*=41.34): N=   1 vendor= 41.14 firm_saving=
0.03 welfare=  41.17 (65% of FB)
   success-fee (monopoly f*=85.59): N=   4 vendor= 50.15 firm_saving=
6.03 welfare=  56.18 (88% of FB)
   success-fee at efficient f=Hk/(k+v)=16.67: N=  21 vendor= 10.59
firm_saving= 52.94 welfare=  63.53

SWE-Verified-like (a=.25,b=.34) | H=100 v=1.0 k=1.0
   first-best: N=  12 surplus=  58.34
   per-attempt (monopoly w*=41.36): N=   1 vendor= 40.36 firm_saving=
0.01 welfare=  40.37 (69% of FB)
   success-fee (monopoly f*=89.59): N=   3 vendor= 48.22 firm_saving=
3.78 welfare=  52.00 (89% of FB)
   success-fee at efficient f=Hk/(k+v)=50.00: N=  12 vendor= 29.17
firm_saving= 29.17 welfare=  58.34

SWE-Verified-like (a=.25,b=.34) | H=300 v=2.0 k=0.5
   first-best: N=  30 surplus= 200.43
   per-attempt (monopoly w*=125.00): N=   1 vendor=124.50 firm_saving=
0.11 welfare= 124.62 (62% of FB)
   success-fee (monopoly f*=263.23): N=   5 vendor=160.17 firm_saving=
16.76 welfare= 176.93 (88% of FB)
   success-fee at efficient f=Hk/(k+v)=60.00: N=  30 vendor= 40.09
firm_saving=160.34 welfare= 200.43
```

```
SWE-Verified-like (a=.25,b=.34) | H=100 v=0.2 k=0.2
   first-best: N=  62 surplus=  72.49
   per-attempt (monopoly w*=42.14): N=   1 vendor= 41.94 firm_saving=
0.04 welfare=  41.97 (58% of FB)
   success-fee (monopoly f*=90.70): N=  12 vendor= 61.52 firm_saving=
5.35 welfare=  66.88 (92% of FB)
   success-fee at efficient f=Hk/(k+v)=50.00: N=  62 vendor= 36.24
firm_saving= 36.24 welfare=  72.49

Synthetic easy (a=.4,b=3) | H=100 v=1.0 k=0.2
   first-best: N=  30 surplus=  43.92
   per-attempt (monopoly w*=4.39): N=   5 vendor= 17.29 firm_saving=
11.89 welfare=  29.19 (66% of FB)
   success-fee (monopoly f*=68.98): N=  10 vendor= 30.35 firm_saving=
7.15 welfare=  37.50 (85% of FB)
   success-fee at efficient f=Hk/(k+v)=16.67: N=  30 vendor=  7.32
firm_saving= 36.60 welfare=  43.92

Synthetic easy (a=.4,b=3) | H=100 v=1.0 k=1.0
   first-best: N=  17 surplus=  33.50
   per-attempt (monopoly w*=6.40): N=   3 vendor= 14.49 firm_saving=
5.87 welfare=  20.36 (61% of FB)
   success-fee (monopoly f*=73.99): N=   8 vendor= 25.25 firm_saving=
4.98 welfare=  30.23 (90% of FB)
   success-fee at efficient f=Hk/(k+v)=50.00: N=  17 vendor= 16.75
firm_saving= 16.75 welfare=  33.50

Synthetic easy (a=.4,b=3) | H=300 v=2.0 k=0.5
   first-best: N=  45 surplus= 151.97
   per-attempt (monopoly w*=14.13): N=   5 vendor= 56.22 firm_saving=
35.87 welfare=  92.09 (61% of FB)
   success-fee (monopoly f*=222.86): N=  13 vendor=108.09 firm_saving=
21.49 welfare= 129.58 (85% of FB)
   success-fee at efficient f=Hk/(k+v)=60.00: N=  45 vendor= 30.39
firm_saving=121.57 welfare= 151.97

Synthetic easy (a=.4,b=3) | H=100 v=0.2 k=0.2
   first-best: N=  97 surplus=  62.27
   per-attempt (monopoly w*=4.04): N=   7 vendor= 20.80 firm_saving=
16.98 welfare=  37.78 (61% of FB)
   success-fee (monopoly f*=84.29): N=  29 vendor= 49.65 firm_saving=
6.72 welfare=  56.36 (91% of FB)
   success-fee at efficient f=Hk/(k+v)=50.00: N=  97 vendor= 31.14
firm_saving= 31.14 welfare=  62.27
```

G.2 Robustness over random parameter draws

Output of `results_robustness.txt`.

```
Random sweep: 296 parameter draws (a~U(.05,.6), b log-U(.1,4), H log-
U(30,500), v/H log-U(.001,.03), k/v log-U(.1,3))
Vendor profit: success-fee >= per-attempt in 99% of draws
Firm saving:   success-fee >= per-attempt in 67% of draws
Welfare/FB:    per-attempt median 0.65 [IQR 0.55-0.77];  success-fee
median 0.91 [IQR 0.88-0.93]
Success-fee welfare > per-attempt welfare in 100% of draws
Median attempts: first-best 28, per-attempt monopoly 1, success-fee
monopoly 6
Efficient-fee vendor share of surplus equals k/(k+v): max abs diff
0.000

Regret from over-estimating vendor quality (H=100, v=1, k=0.2,
b=0.39): firm saving vs human-only when it sets N from believed 'a'
but true 'a' is lower
a_true   contract         saving(correct belief a=0.25 vs true)
saving(believed a=0.25)   loss
 0.25    per-attempt w=2.0         50.71      50.71      0.00
(N_believed=8, N_true=8)
 0.25    success fee f=Hk/(k+v)    50.92      50.92      0.00
(N_believed=21, N_true=21)
 0.15    per-attempt w=2.0         32.48      31.38      1.10
(N_believed=8, N_true=5)
 0.15    success fee f=Hk/(k+v)    34.54      33.59      0.95
(N_believed=21, N_true=12)
 0.10    per-attempt w=2.0         21.12      17.63      3.49
(N_believed=8, N_true=3)
 0.10    success fee f=Hk/(k+v)    23.60      20.60      3.00
(N_believed=21, N_true=8)
 0.05    per-attempt w=2.0          8.78      -0.24      9.02
(N_believed=8, N_true=2)
 0.05    success fee f=Hk/(k+v)    10.93       3.10      7.83
(N_believed=21, N_true=4)
```

G.3 Private vendor quality: representative case

Output of `results_asymmetric.txt`.

```
(i) Profit order for a common attempt budget N (b=.35, H=100, k=.2):
per-attempt w=10  /  success fee f=50
  N=  1: A: low   9.80 high   9.80 | B: low   9.10 high  20.63
  N=  3: A: low  25.31 high  20.34 | B: low  12.33 high  27.34
  N= 10: A: low  73.75 high  46.51 | B: low  14.81 high  32.60
  N= 30: A: low 199.40 high 101.91 | B: low  15.10 high  35.42

(ii) Full-information optimum  high: profit 49.48 via B price 85.3 N=4
low: profit 18.61 via B price 81.9 N=2
     High type if believed low: profit 41.29 (B price 81.9, N=2)

Separating (Riley-type): NO linear contract satisfies both the low
type's no-mimicry and the high type's participation-in-separation
```

```
constraints
Pooling benchmark: expected-profit-maximizing single contract B price
82.6, N=3, profits high 45.44 low 20.70

Welfare (surplus vs human-only) by true type:
  full info:  high 55.58  low 20.98
  separating: high 48.88  low 20.98
  pooling:    high 53.02  low 22.59
  firm saving (H - cost): full info high 6.10, low 2.37; separating
high 7.59; pooling high 7.58 low 1.89
Sweep: separating linear contract exists in 46 of 60 draws (success-
fee type in 46)
```

G.4 Private vendor quality: sweep

Output of `results_asym_sweep.txt`.

```
draws 60; separating exists 46; median distortion 11.0%; low would
mimic FI contract in 98%
welfare sep/FI median 1.07; share higher 74%; firm better off in 74%;
N_sep>=N_FI in 76%
```

G.5 Differentiated-Bertrand equilibria

Output of `results_diff_bertrand.txt`.

```
RepairBench-like (a=.12,b=.39), H=100 v=1 k=0.2; first-best N=10
S^FB^=33.9

 sigma = 1.0
  n | form |  price   N  coverage  firm saving  vendor profit(per
firm)  total vendor profit | welfare/FB | deviation to other form
profitable?
   1 |  A   |   19.60   1    0.95       2.93       19.40
18.416       |  0.632    | yes (dev 25.251 vs eq 18.416)
   1 |  B   |   78.96   3    0.99       4.47       25.54
25.251       |  0.878    | no  (dev 18.416 vs eq 25.251)
   2 |  A   |    0.40   9    1.00      32.57        1.25
1.247        |  0.999    | yes (dev 0.662 vs eq 0.623)
   2 |  B   |    8.02  11    1.00      31.88        1.94
1.943        |  0.999    | no  (dev 0.831 vs eq 0.972)
   3 |  A   |    0.40   9    1.00      32.57        1.25
1.247        |  0.999    | yes (dev 0.421 vs eq 0.416)
   3 |  B   |    7.01  11    1.00      32.31        1.52
1.515        |  0.999    | no  (dev 0.492 vs eq 0.505)
   5 |  A   |    0.40   9    1.00      32.57        1.25
1.247        |  0.999    | yes (dev 0.250 vs eq 0.249)
   5 |  B   |    6.41  11    1.00      32.57        1.26
1.259        |  0.999    | no  (dev 0.251 vs eq 0.252)
  10 |  A   |    0.40   9    1.00      32.57        1.25
1.247        |  0.999    | yes (dev 0.126 vs eq 0.125)
  10 |  B   |    6.01  11    1.00      32.74        1.09
```

```
1.087       |  0.999    | no  (dev 0.108 vs eq 0.109)

 sigma = 3.0
  n | form |  price   N  coverage  firm saving  vendor profit(per
firm)  total vendor profit | welfare/FB | deviation to other form
profitable?
   1 |  A   |   17.80   1    0.83       4.73       17.60
14.585       |  0.587    | yes (dev 22.282 vs eq 14.585)
   1 |  B   |   70.74   4    0.92       7.18       24.32
22.282       |  0.878    | no  (dev 14.585 vs eq 22.282)
   2 |  A   |    0.80   7    1.00      30.44        3.02
3.020        |  0.989    | yes (dev 1.858 vs eq 1.510)
   2 |  B   |   17.43  10    1.00      27.89        5.97
5.966        |  1.000    | no  (dev 2.321 vs eq 2.983)
   3 |  A   |    0.80   7    1.00      30.44        3.02
3.020        |  0.989    | yes (dev 1.169 vs eq 1.007)
   3 |  B   |   13.83  10    1.00      29.41        4.45
4.449        |  1.000    | no  (dev 1.248 vs eq 1.483)
   5 |  A   |    0.80   7    1.00      30.44        3.02
3.020        |  0.990    | yes (dev 0.686 vs eq 0.604)
   5 |  B   |   12.42  10    1.00      30.00        3.86
3.860        |  1.000    | no  (dev 0.678 vs eq 0.772)
  10 |  A   |    0.80   7    1.00      30.44        3.02
3.020        |  0.990    | yes (dev 0.339 vs eq 0.302)
  10 |  B   |   11.22  11    1.00      30.51        3.31
3.312        |  0.999    | no  (dev 0.295 vs eq 0.331)

 sigma = 10.0
  n | form |  price   N  coverage  firm saving  vendor profit(per
firm)  total vendor profit | welfare/FB | deviation to other form
profitable?
   1 |  A   |   21.40   1    0.53       1.13       21.20
11.198       |  0.547    | yes (dev 16.363 vs eq 11.198)
   1 |  B   |   62.32   5    0.73      10.15       22.29
16.363       |  0.866    | no  (dev 11.198 vs eq 16.363)
   2 |  A   |    3.60   3    0.95      21.62        8.39
7.936        |  0.904    | yes (dev 5.951 vs eq 3.968)
   2 |  B   |   45.69   7    0.91      16.42       17.04
15.535       |  0.972    | no  (dev 5.144 vs eq 7.767)
   3 |  A   |    2.40   4    0.97      24.59        6.91
6.714        |  0.947    | yes (dev 3.188 vs eq 2.238)
   3 |  B   |   37.27   8    0.96      19.74       13.95
13.337       |  0.990    | no  (dev 3.027 vs eq 4.446)
   5 |  A   |    2.40   4    0.98      24.59        6.91
6.790        |  0.953    | yes (dev 1.868 vs eq 1.358)
   5 |  B   |   29.06   9    0.98      23.07       10.75
10.545       |  0.998    | no  (dev 1.528 vs eq 2.109)
  10 |  A   |    2.40   4    0.99      24.59        6.91
6.847        |  0.958    | yes (dev 0.929 vs eq 0.685)
  10 |  B   |   29.06   9    0.99      23.07       10.75
```

```
10.649       |  0.999    | no  (dev 0.784 vs eq 1.065)

 sigma = 30.0
  n | form |  price   N  coverage  firm saving  vendor profit(per
firm)  total vendor profit | welfare/FB | deviation to other form
profitable?
   1 |  A   |   22.40   1    0.50       0.13       22.20
11.124       |  0.757    | yes (dev 13.720 vs eq 11.124)
   1 |  B   |   78.96   3    0.54       4.47       25.54
13.720       |  0.871    | no  (dev 11.124 vs eq 13.720)
   2 |  A   |   22.40   1    0.67       0.13       22.20
14.821       |  0.809    | yes (dev 9.434 vs eq 7.411)
   2 |  B   |   70.74   4    0.72       7.18       24.32
17.450       |  0.936    | no  (dev 6.809 vs eq 8.725)
   3 |  A   |   22.40   1    0.75       0.13       22.20
16.668       |  0.835    | yes (dev 7.220 vs eq 5.556)
   3 |  B   |   70.74   4    0.79       7.18       24.32
19.263       |  0.950    | no  (dev 4.906 vs eq 6.421)
   5 |  A   |   22.40   1    0.83       0.13       22.20
18.513       |  0.861    | yes (dev 4.913 vs eq 3.703)
   5 |  B   |   70.74   4    0.86       7.18       24.32
21.010       |  0.962    | no  (dev 3.147 vs eq 4.202)
  10 |  A   |   22.40   1    0.91       0.13       22.20
20.190       |  0.888    | yes (dev 2.733 vs eq 2.019)
  10 |  B   |   62.32   5    0.93      10.15       22.29
20.810       |  0.983    | no  (dev 1.524 vs eq 2.081)
```

G.6 Differentiated-Bertrand sweep

Output of `results_diff_sweep.txt`.

```
Sweep: 240 (parameter draw x n x differentiation) cells; sigma =
{0.03,0.1,0.3} x first-best surplus; n in {2,5}
Fixed-point accuracy within two price-grid steps in 100% of cells
Per-attempt equilibrium NOT deviation-proof (a vendor gains by
switching to a success fee): 71% of cells
Success-fee equilibrium deviation-proof against a switch to per-
attempt: 76% of cells
  n=2 sigma=0.03S: welfare/FB per-attempt 1.00, success fee 0.99;
budget/FB-depth A 0.79, B 1.35; vendor profit B>=A in 80%; firm saving
A>B in 90%  (cells 40)
  n=2 sigma=0.10S: welfare/FB per-attempt 0.98, success fee 1.00;
budget/FB-depth A 0.57, B 1.17; vendor profit B>=A in 98%; firm saving
A>B in 98%  (cells 40)
  n=2 sigma=0.30S: welfare/FB per-attempt 0.90, success fee 0.98;
budget/FB-depth A 0.29, B 0.84; vendor profit B>=A in 100%; firm
saving A>B in 95%  (cells 40)
  n=5 sigma=0.03S: welfare/FB per-attempt 1.00, success fee 0.99;
budget/FB-depth A 0.84, B 1.38; vendor profit B>=A in 55%; firm saving
A>B in 68%  (cells 40)
  n=5 sigma=0.10S: welfare/FB per-attempt 0.99, success fee 1.00;
```

```
budget/FB-depth A 0.64, B 1.29; vendor profit B>=A in 98%; firm saving
A>B in 92%  (cells 40)
  n=5 sigma=0.30S: welfare/FB per-attempt 0.95, success fee 1.00;
budget/FB-depth A 0.41, B 1.00; vendor profit B>=A in 100%; firm
saving A>B in 95%  (cells 40)
```

G.7 Numerical checks of the proofs

Output of `results_verify_proofs.txt`.

```
== Proposition 4 (integer budget): verification ==
293 draws with N^FB^ >= 1
identity R(n) - L(n-1) = S(n-1)(phi_(n-1) q_(n-1) - k): violations 0
sign(R(n) - L(n-1)) = sign(phi_(n-1) - f^e^): violations 0
chain R(n) < R(N^FB^) for all n > N^FB^: violations 0
vendor's best step n_hat = argmax R(n): strictly below N^FB^ in 289
draws, equal to N^FB^ in 4 draws; argmax/brute-force mismatches 0

== Proposition 6(b): convergence in sigma (RepairBench-like, H=100,
v=1, k=0.2, n=5) ==
G_bar <= S^FB^ = 33.9;  rho = exp(G_bar/sigma) - 1;  coverage limit
n/(n+1) = 0.833
 sigma | rho  | bound (1-rho)/(1+rho) | A: Pi(p_hat)/Pi_m  N  coverage
| B: Pi(p_hat)/Pi_m  N  coverage | monopoly N_A N_B
     30 |  2.09 |  0.000               | 1.000          1   0.834    |
0.952          4   0.864    |   1   3
    100 |  0.40 |  0.426               | 1.000          1   0.834    |
1.000          3   0.839    |   1   3
    300 |  0.12 |  0.787               | 1.000          1   0.833    |
1.000          3   0.835    |   1   3
   1000 |  0.03 |  0.933               | 1.000          1   0.833    |
1.000          3   0.834    |   1   3
   3000 |  0.01 |  0.978               | 1.000          1   0.833    |
1.000          3   0.834    |   1   3
```

G.8 Distribution-free checks (Propositions 1, 5, 6)

Output of `results_general_theory.txt`.

```
200 random heterogeneity distributions (finite mixtures with 2-5
atoms, 40% with an unsolvable atom p=0)
(1) q_n nonincreasing: violations 0
(2) Pi(w,f) <= R(N) - (w/v) G_min(N): 0 violations in 13690 contracts
(3) increment formula R(N+1)-R(N) = S(N)[q_N H-(k+v)] - v P(N)(1/q_N -
1/q_(N-1)): violations 0
(4) vendor's grid-optimal contract has per-attempt fee within one grid
step of 0, or induces a single attempt (where G_min=0 and the forms
tie), in 177 of 181 distributions; of the exceptions to w=0, 9 are
single-attempt ties and 4 are other
(5) w=k implements first-best depth under every distribution:
violations 0
```

```
(6) grid-optimal profit never exceeds max_N R(N) (violations 0);
relative gap to max_N R(N): median 0.38%, max 16.83% (grid resolution)
```

Appendix H. Empirical Analyses of Repair Data

H.1 RepairBench (Section 6)

Output of `results_empirical.txt`.

```
Fitted (model x dataset) combos: 84  (n=10 samples/bug, >=80 bugs)
Beta-Binomial beats Binomial by AIC: 84/84;  held-out loglik: 84/84
Zero-inflated beats Beta-Binomial by AIC: 13/84;  held-out loglik:
36/84
Median fraction of bugs with 0/10 successes: 0.59;  median mean
success rate: 0.222
Median fitted (a, b) Beta-Binomial: a=0.12, b=0.39
Median ZI: a=0.21, b=0.41, pi=0.20

== Reservation threshold N* by H/c (median across combos) ==
H/c   N*_BB(median)  N*_ZI(median)  [N*_BB IQR]
   5         1             1   [0, 1]
  10         1             1   [1, 1]
  30         3             3   [2, 5]
 100        11            10   [8, 16]
 300        34            26   [24, 48]

== Held-out cost (units of c) per bug, caps restricted to 0..10
(exact, from real outcomes) ==
H/c  policy                          mean cost over combos   %vs
human-only   best-in-hindsight
   3  human-only                           3.000               0.0%
   3  fixed k=1                            3.344             -11.5%
   3  fixed k=3                            4.602             -53.4%
   3  fixed k=5                            5.873             -95.8%
   3  fixed k=10                           8.953            -198.4%
   3  BB reservation (train fit)           2.962               1.3%
   3  ZI reservation (train fit)           2.962               1.3%
   3  hindsight best fixed                 2.949               1.7%
   3  train-best fixed                     2.962               1.3%

   5  human-only                           5.000               0.0%
   5  fixed k=1                            4.907               1.9%
   5  fixed k=3                            5.998             -20.0%
   5  fixed k=5                            7.198             -44.0%
   5  fixed k=10                          10.190            -103.8%
   5  BB reservation (train fit)           4.772               4.6%
   5  ZI reservation (train fit)           4.787               4.3%
   5  hindsight best fixed                 4.691               6.2%
   5  train-best fixed                     4.773               4.5%

   8  human-only                           8.000               0.0%
```

```
   8  fixed k=1                          7.251                9.4%
   8  fixed k=3                          8.091               -1.1%
   8  fixed k=5                          9.186              -14.8%
   8  fixed k=10                        12.046              -50.6%
   8  BB reservation (train fit)         7.214                9.8%
   8  ZI reservation (train fit)         7.215                9.8%
   8  hindsight best fixed               7.153               10.6%
   8  train-best fixed                   7.216                9.8%

  10  human-only                        10.000                0.0%
  10  fixed k=1                          8.814               11.9%
  10  fixed k=3                          9.487                5.1%
  10  fixed k=5                         10.511               -5.1%
  10  fixed k=10                        13.283              -32.8%
  10  BB reservation (train fit)         8.781               12.2%
  10  ZI reservation (train fit)         8.781               12.2%
  10  hindsight best fixed               8.728               12.7%
  10  train-best fixed                   8.781               12.2%

== Selected models: fits (all bugs) ==
model/dataset                           nbugs  mean_succ  frac0   a
b     N*_BB(H/c=30)  N*_BB(100)  pi_ZI  N*_ZI(30)  N*_ZI(100)
claude-3-7-sonnet-20250219/defects4j      484     0.478   0.37   0.15
0.17         5         15   0.25       4        11
claude-3-7-sonnet-20250219/gitbugjava      90     0.234   0.63   0.08
0.26         3          8   0.35       2         7
deepseek-v3/defects4j                     482     0.396   0.48   0.09
0.13         3          9   0.00       3         9
deepseek-v3/gitbugjava                     90     0.223   0.67   0.06
0.21         2          6   0.06       2         6
gpt-4o-2024-11-20/defects4j               484     0.350   0.49   0.12
0.23         4         12   0.23       4        11
gpt-4o-2024-11-20/gitbugjava               90     0.198   0.68   0.07
0.29         2          7   0.48       2         6
llama-3.3-70b-instruct/defects4j          484     0.234   0.56   0.14
0.46         4         14   0.16       4        13
llama-3.3-70b-instruct/gitbugjava          90     0.171   0.71   0.07
0.39         2          7   0.67       2         5
o4-mini-2025-04-16-high/defects4j         484     0.538   0.27   0.25
0.23         8         25   0.20       6        14
o4-mini-2025-04-16-high/gitbugjava         90     0.313   0.44   0.16
0.33         5         16   0.00       5        16
qwen-2.5-coder-32b-instruct/defects4j     479     0.271   0.53   0.14
0.40         4         14   0.28       4        12
qwen-2.5-coder-32b-instruct/gitbugjava     90     0.138   0.66   0.13
0.73         3         12   0.00       3        12
```

H.2 SWE-bench logs and published coverage points

Output of `results_swebench.txt`.

```
== (A) Calibration from published coverage points (aggregate; SWE-
bench Verified=500 tasks unless noted) ==
source                                    pass@1  pass@k   k     a
b     N*(H/c=30) N*(100) N*(300) N*(1000)
DeepSWE-Preview (Qwen3-32B), Verified     0.422  0.710   16  0.249
0.341        7      25      75     249
    implied coverage: @1=0.42, @2=0.51, @4=0.59, @8=0.66, @16=0.71,
@32=0.76, @64=0.79, @128=0.83, @250=0.85
Cortexa o3, Verified (pass@7=75%)         (pass@1 not stated in repo
README; skipped)
Large Language Monkeys, Lite, DS-Coder-V2  0.159  0.560  250  0.122
0.645        3      12      36     122
    implied coverage: @1=0.16, @2=0.22, @4=0.28, @8=0.33, @16=0.39,
@32=0.43, @64=0.48, @128=0.52, @250=0.56

== (B) Cross-submission instance heterogeneity (proxy; systems differ,
not iid samples) ==
lite: 84 submissions; instances resolved by >=1 submission: 265/300;
never resolved: 35 (11.7%)
   share-of-systems resolving an instance (resolved-by-some instances)
percentiles 10/25/50/75/90: [0.01 0.07 0.36 0.63 0.82]
   Beta fit across systems: a=0.44, b=0.97 (U-shaped if both<1),
mean=0.31
verified: 135 submissions; instances resolved by >=1 submission:
471/500; never resolved: 29 (5.8%)
   share-of-systems resolving an instance (resolved-by-some instances)
percentiles 10/25/50/75/90: [0.07 0.31 0.61 0.81 0.91]
   Beta fit across systems: a=0.69, b=0.77 (U-shaped if both<1),
mean=0.47
```

H.3 HumanEvalFix with 40 samples per bug

Output of `results_hef.txt`.

```
=== google_gemma-3-4b-it: 164 bugs x 40 samples; mean success 0.514;
share with 0/40: 0.32; share with 40/40: 0.38
    Beta-binomial a=0.097 b=0.088; AIC bin=7293 vs BB=739 vs ZI-BB=740
(ZI: a=0.201 b=0.097 pi=0.207)
    Fit from first 10 samples only: a=0.118 b=0.109
    H/c=  10: N* BB (n=40)=  1  N* BB (first 10)=  1  N* ZI (n=40)=  1
    H/c=  30: N* BB (n=40)=  3  N* BB (first 10)=  4  N* ZI (n=40)=  3
    H/c= 100: N* BB (n=40)= 10  N* BB (first 10)= 12  N* ZI (n=40)=  9
    H/c= 300: N* BB (n=40)= 29  N* BB (first 10)= 36  N* ZI (n=40)= 23
    Held-out cost (units of c) by H/c: human-only | fixed k=1 | k=5 |
k=10 | BB reservation | ZI reservation | best fixed in hindsight
      H/c=  10:   10.0 |   5.74 |   6.34 |   7.79 |   5.74 (N=1) |
5.74 (N=1) |   5.73 (k=2)
      H/c=  30:   30.0 |  15.21 |  13.68 |  14.49 |  13.83 (N=3) |
13.83 (N=3) |  13.68 (k=5)
      H/c= 100:  100.0 |  48.38 |  39.35 |  37.97 |  37.97 (N=10) |
38.14 (N=8) |  37.97 (k=10)
```

```
    Coverage at j=1,5,10,20,40  observed: 0.514, 0.622, 0.650, 0.667,
0.677   BB first-10 fit: 0.519, 0.614, 0.645, 0.674, 0.700   ZI first-
10 fit: 0.519, 0.614, 0.645, 0.674, 0.700

=== meta-llama_llama-3.2-3b-instruct: 164 bugs x 40 samples; mean
success 0.437; share with 0/40: 0.07; share with 40/40: 0.09
    Beta-binomial a=0.504 b=0.581; AIC bin=4320 vs BB=1180 vs ZI-
BB=1182 (ZI: a=0.504 b=0.581 pi=0.000)
    Fit from first 10 samples only: a=0.536 b=0.643
    H/c=  10: N* BB (n=40)=  4  N* BB (first 10)=  5  N* ZI (n=40)=  4
    H/c=  30: N* BB (n=40)= 15  N* BB (first 10)= 15  N* ZI (n=40)= 15
    H/c= 100: N* BB (n=40)= 50  N* BB (first 10)= 53  N* ZI (n=40)= 50
    H/c= 300: N* BB (n=40)=151  N* BB (first 10)=160  N* ZI (n=40)=151
    Held-out cost (units of c) by H/c: human-only | fixed k=1 | k=5 |
k=10 | BB reservation | ZI reservation | best fixed in hindsight
      H/c=  10:   10.0 |   6.65 |   5.68 |   6.15 |   5.68 (N=5) |
5.68 (N=5) |   5.68 (k=4)
      H/c=  30:   30.0 |  17.95 |  11.62 |  10.45 |  10.31 (N=17) |
10.31 (N=17) |  10.30 (k=15)
      H/c= 100:  100.0 |  57.50 |  32.40 |  25.50 |  16.36 (N=40) |
16.36 (N=40) |  16.36 (k=40)
    Coverage at j=1,5,10,20,40  observed: 0.437, 0.730, 0.817, 0.876,
0.927   BB first-10 fit: 0.454, 0.733, 0.812, 0.869, 0.909   ZI first-
10 fit: 0.454, 0.733, 0.812, 0.869, 0.909

=== mistralai_ministral-8b-2512: 164 bugs x 40 samples; mean success
0.858; share with 0/40: 0.04; share with 40/40: 0.60
    Beta-binomial a=0.482 b=0.090; AIC bin=3653 vs BB=656 vs ZI-BB=651
(ZI: a=0.934 b=0.109 pi=0.039)
    Fit from first 10 samples only: a=0.573 b=0.097
    H/c=  10: N* BB (n=40)=  5  N* BB (first 10)=  6  N* ZI (n=40)=  4
    H/c=  30: N* BB (n=40)= 14  N* BB (first 10)= 17  N* ZI (n=40)=  8
    H/c= 100: N* BB (n=40)= 48  N* BB (first 10)= 57  N* ZI (n=40)= 16
    H/c= 300: N* BB (n=40)=145  N* BB (first 10)=172  N* ZI (n=40)= 29
    Held-out cost (units of c) by H/c: human-only | fixed k=1 | k=5 |
k=10 | BB reservation | ZI reservation | best fixed in hindsight
      H/c=  10:   10.0 |   2.43 |   2.34 |   2.73 |   2.65 (N=9) |
2.65 (N=9) |   2.23 (k=3)
      H/c=  30:   30.0 |   5.29 |   4.13 |   4.43 |   5.76 (N=27) |
5.76 (N=27) |   4.12 (k=4)
      H/c= 100:  100.0 |  15.29 |  10.42 |  10.37 |  12.60 (N=40) |
12.60 (N=40) |  10.31 (k=7)
    Coverage at j=1,5,10,20,40  observed: 0.858, 0.937, 0.950, 0.956,
0.957   BB first-10 fit: 0.856, 0.945, 0.963, 0.975, 0.983   ZI first-
10 fit: 0.865, 0.942, 0.951, 0.956, 0.958
```